

Paper OOL-6: Autocatalytic Closure and Chemical Memory: When Reaction Networks Begin to Remember

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Abstract

This paper isolates autocatalytic closure from later claims about alphabets, homochirality, or protocells. A reaction network begins to remember only when its products help regenerate the conditions that make them. In CDFD terms, this is the first serious biological use of M_s : retained organization through feedback. The companion script performs a conservative closure scan and reports sub-ignition persistence enhancement rather than completed metabolic ignition.

1 Introduction

Papers 1–5 supplied energy, transport, compartments, and selective interfaces. None of these alone guarantees *self-reinforcing* chemistry. Autocatalysis and cross-catalysis supply positive feedback: products accelerate their own formation [1, 2]. In open systems, however, diffusion and dilution supply negative feedback. Closure is the balance point.

Gap from Paper 5: catalytic selection must escalate into network persistence.

Status: Inferred

Capstone channel mapping. Metabolic closure is not single-channel: losing input Φ , electron throughput σ_e , proton coupling σ_p , or stabilization capacity against S collapses the bundled J in the weakest-link sense even when Γ inequalities locally favor amplification.

2 Model and Assumptions

2.1 Network mass balances

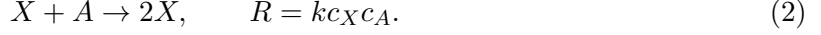
For species X_i with concentrations c_i ,

$$\frac{\partial c_i}{\partial t} = \sum_j R_{ij}(\mathbf{c}) - \nabla \cdot \mathbf{J}_i + S_i, \quad \mathbf{J}_i = -\frac{1}{C_i} \nabla c_i, \quad (1)$$

with sources S_i . Here C_i is a *transport resistance* for species i (not to be confused with product concentration symbols in legacy drafts).

Status: Derived

2.2 Minimal autocatalytic core



Well-mixed loss to a compartment of scale L scales as $D_{\text{eff}}c_X/L^2$ with $D_{\text{eff}} \sim 1/C$ in the resistance picture. Then

$$\frac{dc_X}{dt} = kc_Xc_A - \frac{1}{CL^2}c_X = c_X \left(kc_A - \frac{1}{CL^2} \right). \quad (3)$$

Status: Derived

2.3 Ignition parameter

Define

$$\Gamma = kc_A CL^2. \quad (4)$$

If $\Gamma > 1$, exponential growth of c_X results in this linearized well-mixed limit; if $\Gamma < 1$, trajectories decay toward zero without other sources. The boundary $\Gamma = 1$ is a mean-field caricature; spatial patterning and nonlinear saturation modify bifurcations.

Status: Derived

2.4 Vector closure

A network is *closed* if every required intermediate is produced internally faster than it leaks, $\forall i, \sum_j k_{ij}c_j > (C_i L_i^2)^{-1}$ in the same schematic sense.

Status: Hypothesis

2.5 Energy coupling

Rates may scale with coupled fluxes $R_i \propto |\mathbf{J}_e \cdot \mathbf{J}_p|$ when redox drive gates chemistry [3]. We do not specify constants.

Status: Hypothesis

2.6 Linear stability (Jacobian)

Let $\mathbf{J}_{\text{jac}}$ be the Jacobian of the reaction vector field. Local stability of a steady state requires $\text{Re } \lambda < 0$ for all eigenvalues λ of $\mathbf{J}_{\text{jac}}$. Autocatalysis can instead produce unstable modes, limit cycles, or blow-up depending on constraints and saturation.

Status: Derived

3 Main Results

3.1 Conceptual

Metabolic organization begins when positive feedback loops beat transport losses [4]. Compartments increase C (resistance to escape), improving Γ for fixed k, c_A, L .

Status: Inferred

4 Numerical Verification

`supplementary_ool_06.py` integrates a reduced two-parameter caricature. Reported final Φ values:

- $(\Phi_0 = 0.1, C = 1.0) \rightarrow 0.073$
- $(\Phi_0 = 0.5, C = 1.0) \rightarrow 0.554$
- $(\Phi_0 = 0.1, C = 5.0) \rightarrow 0.102$
- $(\Phi_0 = 0.5, C = 5.0) \rightarrow 0.849$

Higher constraint C raises the final state relative to comparable baselines, but **no run exhibits runaway ignition** in the companion script. Narratives about “critical metabolism” must therefore remain Hypothesis/Open until bifurcation scans identify supercritical parameters.

Status: Derived

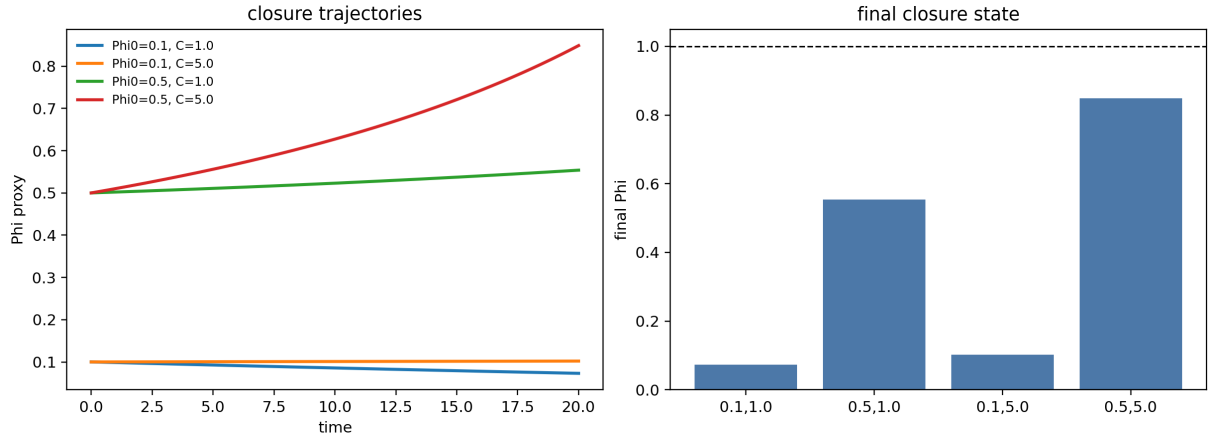


Figure 1: Autocatalytic-closure diagnostic. The trajectories show persistence enhancement under stronger constraint but do not prove completed metabolic ignition. The figure is a disciplined threshold check, not a claim of living metabolism.

5 Interpretation

The mathematics clarifies what must be proven: closure is a competition between amplification and loss. The numerics currently demonstrate *sub-ignition* persistence enhancement, not autonomous self-acceleration to a living steady state.

Status: Inferred

6 Limitations and Open Problems

- Mismatch between rhetorical “ignition” and outputs is explicitly documented.
- Need bifurcation analysis in (k, c_A, C, L) space with spatial resolution.
- Saturation chemistry and side reactions omitted.

Status: Open

7 Conclusion

This paper states the closure criterion clearly and shows that the companion script does not yet realize full ignition. Paper 7 moves to aromatic carriers where pattern persistence becomes an information-relevant quantity.

Status: Inferred

8 Reproducibility Note

`python supplementary_o01_06.py`

9 Modern Network Boundary

Autocatalysis is not enough by itself. A closure claim must distinguish three layers: self-amplification of a species, mutual regeneration of a reaction set, and persistence of an organized network through environmental cycling. The modern CDFD reading is therefore strict: closure becomes biological only when it couples to chemical memory, boundary retention, and later copying. This is why Paper 6 ends before homochirality or genetic inheritance are claimed.

Status: Derived guardrail

References

- [1] Manfred Eigen. Selforganization of matter and the evolution of biological macromolecules. *Naturwissenschaften*, 58:465–523, 1971.
- [2] Stuart A. Kauffman. Autocatalytic sets of proteins. *Journal of Theoretical Biology*, 119:1–24, 1986.
- [3] Nick Lane, John F. Allen, and William Martin. How did LUCA make a living? chemiosmosis in the origin of life. *BioEssays*, 32:271–280, 2010.
- [4] John Maynard Smith and Eörs Szathmáry. *The Major Transitions in Evolution*. Oxford University Press, 1997.